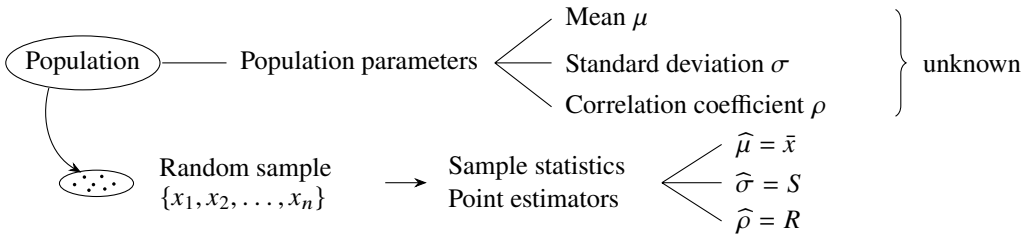
STAT A411/611 Lecture Notes

Instructor: Rieken Venema **Office:** SSB 158E
Office hours: MWF 12:00–12:45 **Class:** 2:30–3:45

Chapter 7: Estimation

Section 7.1: Point Estimation

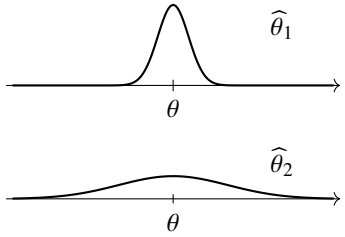


θ	An unknown population parameter.
$\widehat{\theta}$	A point estimator.
$\widehat{\theta} = 3.12$	A point estimate.

Desirable Properties of Point Estimators

1. Small bias: $E[\widehat{\theta}] - \theta$.
2. Small standard deviation: $\mathrm{sd}(\widehat{\theta})$.

If $E[\widehat{\theta}] = \theta$, then $\widehat{\theta}$ is an unbiased point estimator of θ.



Independent and Identically Distributed (I.I.D.)

The random variables $X_1, X_2, \ldots, X_n$ are *I.I.D.* if they are independent and identically distributed. In particular,

$$E[X_i] = \mu, \qquad \mathrm{sd}(X_i) = \sigma, \qquad i = 1, 2, \ldots, n.$$

Property: The Sample Mean Is Unbiased

$\overline{X}$ is an unbiased point estimator of μ.

Proof. Since $E[\,\cdot\,]$ is a linear operator,

$$\begin{aligned} E[\overline{X}] &= E\left[\frac{1}{n}X_1 + \frac{1}{n}X_2 + \cdots + \frac{1}{n}X_n\right] \\ &= \frac{1}{n}E[X_1] + \frac{1}{n}E[X_2] + \cdots + \frac{1}{n}E[X_n] \\ &= \underbrace{\frac{1}{n}\mu + \frac{1}{n}\mu + \cdots + \frac{1}{n}\mu}_{n \text{ times}} = \mu. \end{aligned}$$

Example 7.1.1: A Fair Six-Sided Die

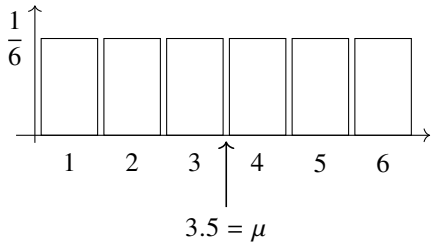
Roll a fair six-sided die. Let X be the side coming up. Then

$$\mathcal{X} = \{1, 2, 3, 4, 5, 6\}, \qquad P(X = 1) = P(X = 2) = \cdots = P(X = 6) = \frac{1}{6},$$

so $X \sim U(6)$.

Using the expected-value formula from Section 3.3,

$$\begin{aligned} E[X] &= \sum_{x=1}^{6} x\, P(X = x) \\ &= 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + \cdots + 6 \cdot \frac{1}{6} \\ &= \frac{21}{6} = 3.5. \end{aligned}$$



Repeated Experiments

Experiment 1: Roll a fair die 30 times. Let $\bar{x}_1$ be the average outcome:

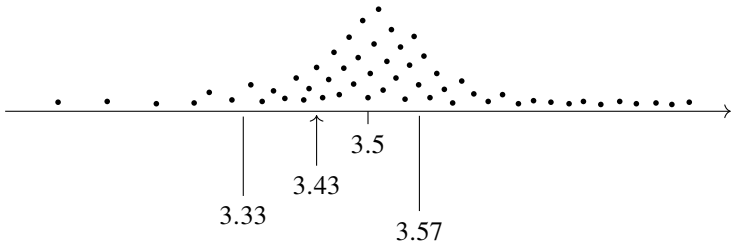
$$\bar{x}_1 = \frac{2 + 2 + 4 + 1 + 1 + 1 + \cdots + 5}{30} \approx 3.43.$$

Repeat the experiment:

$$\begin{aligned} \text{Experiment 2:} \quad & \bar{x}_2 \approx 3.33, \\ \text{Experiment 3:} \quad & \bar{x}_3 = 3.60, \\ & \vdots \\ \text{Experiment 56:} \quad & \bar{x}_{56} \approx 3.57. \end{aligned}$$

Make a dotplot of the 56 sample means.

Figure 7.1



The 56 $\bar{x}$-values are centered around $\mu = 3.5$, illustrating that $\overline{X}$ is an unbiased point estimator of μ.

Properties of Linear Combinations

Let $X_1, \ldots, X_n$ be n random variables with finite variances, and let $a_1, \ldots, a_n$ be n constants.

(a)
$$E\left[\sum_{i=1}^{n} a_i X_i\right] = \sum_{i=1}^{n} a_i E[X_i].$$

$E[\,\cdot\,]$ is a linear operator.

(b)
$$\mathrm{V}\left[\sum_{i=1}^{n} a_i X_i\right] = \sum_{i=1}^{n}\sum_{j=1}^{n} a_i a_j \underbrace{\mathrm{Cov}(X_i, X_j)}_{\text{covariance}}.$$

(c) If the X_i are independent, then
$$\mathrm{Cov}(X_i, X_j) = \begin{cases} 0, & i \neq j, \\ \mathrm{V}[X_i], & i = j. \end{cases}$$

So,
$$\mathrm{V}\left[\sum_{i=1}^{n} a_i X_i\right] = \sum_{i=1}^{n} a_i^2 \mathrm{V}[X_i].$$

Example: Variance and Standard Error of the Sample Mean

Let $X_1, \ldots, X_n$ be an I.I.D. random sample with
$$E[X_i] = \mu, \qquad \mathrm{V}[X_i] = \sigma^2.$$

Then
$$E[\overline{X}] = \mu.$$

Using property (c),
$$\begin{aligned} \mathrm{V}[\overline{X}] &= \sum_{i=1}^{n} \frac{1}{n^2} \mathrm{V}[X_i] \\ &= \sum_{i=1}^{n} \left[\frac{1}{n^2}\sigma^2\right] \\ &= \frac{1}{n}\sigma^2. \end{aligned}$$

So,
$$\mathrm{sd}(\overline{X}) = \sqrt{\frac{\sigma^2}{n}} = \frac{\sqrt{\sigma^2}}{\sqrt{n}} = \frac{\sigma}{\sqrt{n}}.$$

This is the *standard error* of the sample mean $\overline{X}$.

Property: Unbiased Estimators of the Mean and Variance

Let $X_1, X_2, \ldots, X_n$ be an I.I.D. random sample, with $n > 1$, such that

$$E[X_i] = \mu, \qquad \text{sd}(X_i) = \sigma.$$

Then:

(a) The sample mean $\overline{X}$ is an unbiased point estimator of the population mean μ.

(b) The sample variance S^2 is an unbiased point estimator of the population variance σ^2.

Proof

For (a), $E[\overline{X}] = \mu$.

For (b), we want to show that $E[S^2] = \sigma^2$. Using the sample-variance identity from Section 6.3,

$$\begin{aligned} E[S^2] &= E\left[\frac{\sum_{i=1}^n X_i^2 - n\overline{X}^2}{n-1}\right] \\ &= \frac{1}{n-1}\left[\sum_{i=1}^n E[X_i^2] - nE[\overline{X}^2]\right]. \end{aligned} \tag{1}$$

Since

$$\begin{aligned} \mathrm{V}[X_i] &= E[X_i^2] - \big(E[X_i]\big)^2, \\ E[X_i^2] &= \sigma^2 + \mu^2. \end{aligned} \tag{2}$$

Similarly,

$$\begin{aligned} \mathrm{V}[\overline{X}] &= E[\overline{X}^2] - \big(E[\overline{X}]\big)^2, \\ \frac{\sigma^2}{n} &= E[\overline{X}^2] - \mu^2, \\ E[\overline{X}^2] &= \frac{\sigma^2}{n} + \mu^2. \end{aligned} \tag{3}$$

By combining (1), (2), and (3),

$$\begin{aligned} E[S^2] &= \frac{1}{n-1}\left[n(\sigma^2 + \mu^2) - n\left(\frac{\sigma^2}{n} + \mu^2\right)\right] \\ &= \frac{1}{n-1}\left[n\sigma^2 - \sigma^2\right] \\ &= \frac{1}{n-1}\left[(n-1)\sigma^2\right] \\ &= \sigma^2. \qquad \square \end{aligned}$$

Why the Denominator Is $n - 1$

The sample variance is defined as

$$S^2 = \frac{\sum_{i=1}^{n}(X_i - \overline{X})^2}{n-1}, \qquad \text{not} \qquad \frac{\sum_{i=1}^{n}(X_i - \overline{X})^2}{n}.$$

The second expression is the average squared deviation around the sample mean. It is a biased estimator of σ^2 when $\sigma^2 > 0$:

$$\begin{aligned} E\left[\frac{\sum_{i=1}^{n}(X_i - \overline{X})^2}{n}\right] &= E\left[\frac{n-1}{n}S^2\right] \\ &= \frac{n-1}{n}E[S^2] \\ &= \left(1 - \frac{1}{n}\right)\sigma^2 \\ &\neq \sigma^2. \end{aligned}$$

However, $\dfrac{\sum_{i=1}^{n}(X_i - \overline{X})^2}{n}$ is an *asymptotically unbiased* point estimator of σ^2:

$$n \longrightarrow \infty \quad \Longrightarrow \quad E\left[\frac{\sum_{i=1}^{n}(X_i - \overline{X})^2}{n}\right] \longrightarrow \sigma^2.$$

Sample Standard Deviation

$$E[S^2] = \sigma^2 \quad \text{does not imply} \quad E[S] = \sigma.$$

In general, $E[S] \neq \sigma$: S is a biased point estimator of σ.

Homework

2, 4, 6, 8; 10(a, b, c); 12(a, b, c, d, e); 14.

Section 7.2: Methods of Point Estimation

Two statistical methods for finding "optimal" point estimators of population parameters are:

I. The method of moments.

II. The maximum likelihood method.

Maximum likelihood estimators (MLEs) often have stronger statistical properties than method-of-moments estimators (MOMEs), so maximum likelihood is often preferred. The second method can be more complicated.

The desirable properties remain small bias and small standard deviation.

Section 7.3: Sampling Distributions

Property 1

If $X_1, X_2, \ldots, X_n$ are independent random variables with

$$X_i \sim N(\mu_i, \sigma_i^2),$$

and $a_1, a_2, \ldots, a_n$ are n constants, then

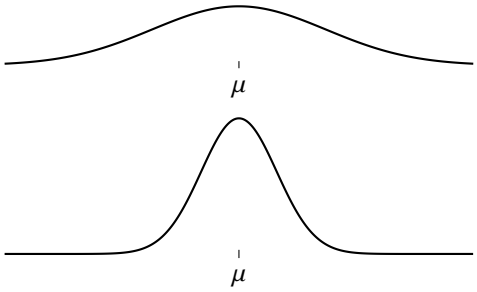
$$\sum_{i=1}^{n} a_i X_i \sim N\left(\sum_{i=1}^{n} a_i \mu_i,\ \sum_{i=1}^{n} a_i^2 \sigma_i^2\right).$$

Property 2

Let $X_1, \ldots, X_n$ be independent with $X_i \sim N(\mu, \sigma^2)$. Then

$$\overline{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

Example



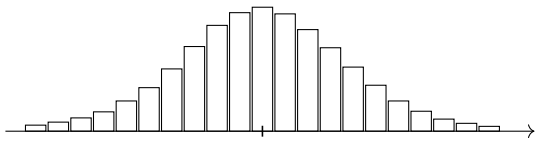
Probability distribution of X_i, $i = 1, 2, \ldots, n$

Probability distribution of $\overline{X}$

Example: Finishing Times in a 10 km Run

The finishers in a 10 km run have finishing times with

$$\bar{x} = 61 \text{ min}, \qquad s = 9 \text{ min}.$$



61 minutes

Model. Let X be the finishing time, in minutes, of a randomly selected finisher:

$$X \sim N(61, 81).$$